

The HL7 Cutover

RAYD stops querying the Philips PACS database directly and moves to a pipeline built on open standards, resolving a vendor compliance block ahead of launch.

PREPARED

10 Sep 2026

APPLIES TO

All current installs

STATUS

Decided, one open item

— THE ISSUE

Philips does not permit direct database access

RAYD's ETL currently connects straight to the Philips PACS Oracle schema (DIDB_STUDIES and related tables) using a stored username and password. Philips's support terms prohibit this, for three reasons, and it affects every site running RAYD today, not one contract.

01 Performance load on production PACS

02 Patient data exposure

03 Proprietary schema access

— THE DECISION

Receive data the way a third party is meant to

Stop querying Oracle live. Take clinical data as HL7 messages Philips already sends, and take the handful of figures HL7 cannot carry as one small aggregated export, with no patient data, computed on their side.

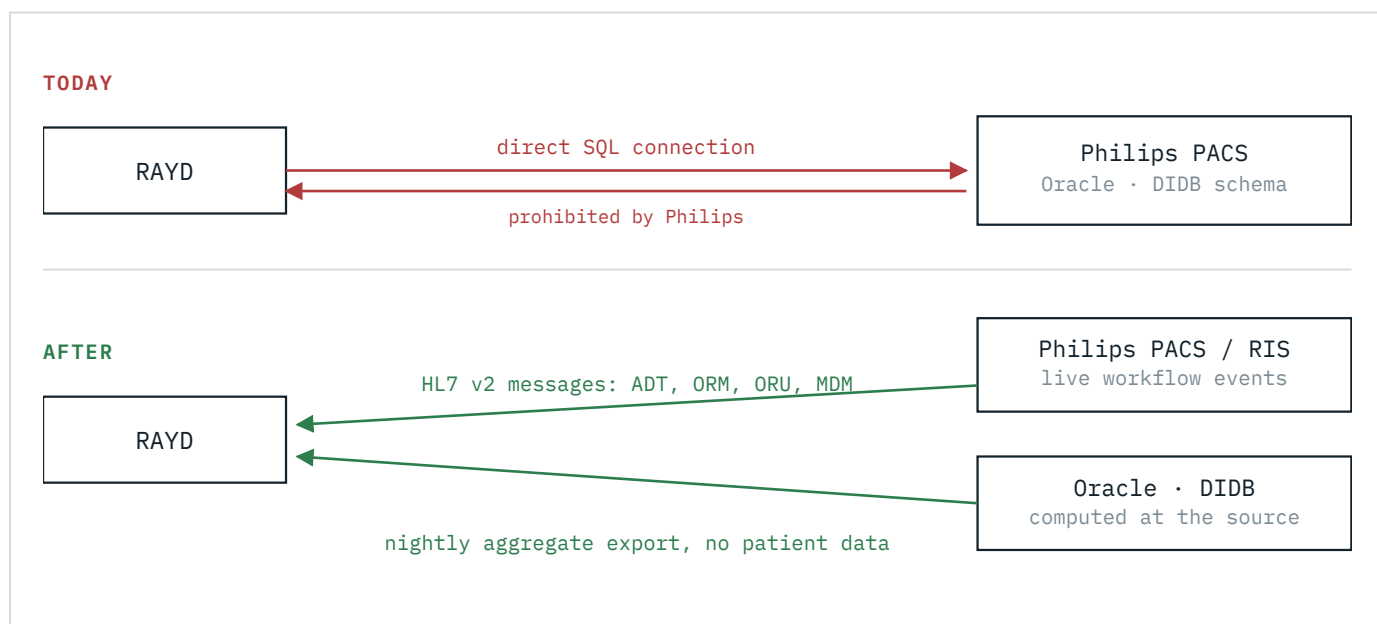


Fig. 1. The arrow reverses: RAYD stops reaching into Oracle, and the source systems push instead.

Every dependency, resolved three ways

KEPT no change to what's reported

Patient demographics, class, location	ADT messages
Referring and reading physicians	ORM and ORU messages
Report text and the prelim, final, addendum chain	MDM messages
Order status	ORM messages
Financial and RVU analytics	already sourced from RIS, untouched
Storage totals (report 29, storage KPIs)	nightly aggregate export

CHANGED recovered through a new mechanism

Device and scanner attribution (reports 25, 34)	new field on the message sent when the scan runs
Turnaround time start point	moves from PACS ingest time to scan time
Image and series counts	added to the nightly aggregate export
Radiologist vs. resident role split	local roster table, kept current by an admin
LAUMC site split (RH and SJH)	patient location code, 1000 and 2000

LOST no substitute

Patient identity merge conflict finder	one small admin screen, not a critical dependency
CD and DVD distribution log (report 30)	separate system, open item below

Each reason Philips gave, answered

PERFORMANCE	No live queries touch production. The one remaining export runs on Philips's own schedule, off their peak hours, computed with their own indexes.
PATIENT SAFETY	No standing database credential lives in RAYD. Clinical data arrives only through the interface Philips already supports, and the export itself carries zero patient data: dates, devices, and totals only.
PROPRIETARY ACCESS	RAYD never sees their schema. HL7 is the standard interoperability interface any third party is expected to use, not a custom grant into Philips's internal tables.

OPEN ITEM, NEEDS A DECISION

Report 30, CD and DVD distribution, turns out to depend on a third Oracle schema (CDSURF, the system Philips uses to burn discs) that no one had flagged before this review. No HL7 message carries a "disc was burned" event either way, so this needs its own scoped export, plus a separate check on whether CDSURF falls under the same restriction as DIDB.

RAYD · Philips PACS integration

Next: define the HL7 interface spec for each site and confirm the export with each PACS administrator